information

MDPI

Article

Information-Driven Team Collaboration in RoboCup Rescue

Abhijot Bedi [1], Shelley Zhang [2,*] and Eugene Chabot [3]

[1] Department of Computer and Information Science, University of Massachusetts Dartmouth, North Dartmouth, MA 02747, USA; abhijotbedi2@gmail.com
[2] School of Science, Technology, and Health, Gordon College, Wenham, MA 01984, USA
[3] NUWC Division Newport, Newport, RI 02841, USA; eugene@uri.edu
* Correspondence: shelley.zhang@gordon.edu

Abstract

Efficient collaboration in multi-robot systems (MRSs) is essential for handling complex tasks in dynamic environments under physical constraints. This study employs the RoboCup Rescue Simulation (RCRS) platform, which supports programmable rescue agents in disaster response scenarios, to investigate collaborative strategies for MRS. The proposed approach integrates a task modeling framework into RCRS to enable systematic task decomposition and coordinated request handling among platoon agents. A dedicated communication protocol further allows agents to share and exploit information dynamically in changing conditions. Experiments demonstrate simulation performance improvements ranging from 12% to 48% over default agents across complex map configurations. Results highlight the effectiveness of structured multi-agent system (MAS) collaboration mechanisms when adapted to practical physical constraints, indicating strong potential for enhancing cooperative performance in real-world multi-robot applications.

Keywords: multi-robot systems; multi-agent system; information-driven collaboration; RoboCup rescue

1. Introduction

Multi-robot systems (MRSs) constitute a central research area in contemporary Artificial Intelligence (AI). Coordinated teams of specialized robots streamline complex tasks and substantially extend human capabilities [1]. MRSs have been successfully applied across diverse domains. Examples include large-scale data collection in aerial, forest, urban, and marine environments [2], Urban Search and Rescue (USAR) operations in hazardous settings [3,4], manufacturing, environmental monitoring, transportation, and infrastructure inspection [5–7].

Deployment of MRSs in large-scale USAR missions, however, faces major challenges. Disaster environments are inherently unpredictable, featuring unstable structures, varied terrain, and hazardous materials that severely complicate navigation and operation [3]. Moreover, destruction of communication infrastructure frequently disrupts standard channels, rendering dynamic task allocation extremely difficult [8].

This paper presents task-scheduling and information-sharing strategies for heterogeneous rescue teams within the RoboCup Rescue Simulation (RCRS) platform [9]. The objective is to bridge theoretical multi-agent system (MAS) coordination mechanisms with the practical constraints of physical MRS, including spatial limitations, physical interactions, and environmental dynamics.

Check for updates

Academic Editor: Leon Sterling

Received: 7 October 2025
Revised: 8 December 2025
Accepted: 16 December 2025
Published: 22 December 2025

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license.

The present work advances RCRS agent capabilities through a task-scheduling framework for Police Force (PF) agents based on the Task Analysis, Environment Modeling, and Simulation (TAEMS) language [10], which enables systematic decomposition and prioritization of blockade-clearing tasks, and through a communication protocol that supports collaboration among Police Force, Ambulance Team (AT), and Fire Brigade (FB) agents using only voice-channel messages, thereby eliminating dependence on radio channels. A preliminary version of the protocol appeared in [11]; the current study provides the first full integration of this protocol with TAEMS scheduling together with comprehensive evaluation.

The paper is organized as follows: Section 2 reviews related work on task allocation and inter-agent collaboration in RCRS. Section 3 describes the RoboCup Rescue Simulation environment. Section 4 introduces the TAEMS modeling language. Section 5 details the integration of TAEMS scheduling and the voice-channel communication protocol. Section 6 presents the experimental setup and performance comparison with baseline agents. Finally, Section 7 discusses results, limitations, and future research directions.

2. Related Work

Significant progress has been achieved in multi-agent systems within the RoboCup Rescue Simulation domain. Lu et al. [12] addressed path planning in RCRS by combining the A* heuristic algorithm with Ant Colony Optimization (ACO) swarm intelligence. This hybrid approach enabled intelligent coordination and blockage clearance under limited perception and absence of global information. Similarly, Lyu et al. [13] proposed an adaptive A* graph routing method that selects paths according to real-time disaster conditions and incorporates a smoothed obstacle-selection strategy, allowing police agents to prioritize blockages most aligned with their current direction of movement. Both works rely on enhanced A*-based path finding with distinct heuristics, whereas the present study emphasizes high-level task prioritization based on gathered information and a local scheduling algorithm.

The 2024 RoboCup Rescue champion team [14] enhanced disaster response by granting agents autonomous decision-making capabilities within assigned zones, followed by communication-driven expansion of search areas. This approach shares the high-level principle of using communication to drive coordination, although it employs different underlying methodologies.

Auction-based mechanisms remain widely applied for task allocation in RCRS. Sedaghat et al. [15] introduced a centralized auction system in which non-police agents request blockage clearance through their centers, with the police center acting as auctioneer. Although effective in certain scenarios, this method imposes centralized coordination, reduces agent-level decentralization, and requires extensive message exchange through centers—an unrealistic demand in post-disaster environments with severely constrained communication infrastructure.

Decentralized coordination in RCRS has frequently been formulated as a Distributed Constraint Optimization Problem (DCOP) [16,17]. Related research modeled task allocation for Ambulance Team and Fire Brigade agents as Coalition Formation with Spatial and Temporal Constraints (CFST) [18]. The CFST formulation explicitly incorporates spatial constraints (agent/task positions) and temporal constraints (deadlines and execution durations). A dedicated DCOP encoding combined with algorithms such as Max-Sum was developed to provide efficient, adaptive decentralized solutions. The present work adopts the TAEMS language to represent and reason about analogous spatial and temporal constraints during rescue operations.

Drew [19] reviewed multi-robot systems in real search-and-rescue applications, stressing the need to integrate machine learning with classical control to improve perception,

coordination, and human–robot interaction. Xueke et al. [20] proposed a cooperative fire-fighting task-planning method in which multiple agents learn joint strategies via the Proximal Policy Optimization (PPO) algorithm after extensive offline training. Although reinforcement learning achieves strong performance, it depends on lengthy pre-training phases. In contrast, the present approach extends agent perception through real-time information sharing, enhancing team coordination without requiring offline training.

Sun et al. [21] surveyed multi-agent coordination across diverse domains, noting that coordination often draws inspiration from swarm intelligence to produce emergent behavior. The survey identifies persistent challenges in scalability, heterogeneity, and maturity of distributed learning, while advocating hybrid coordination schemes for large-scale MAS. These recommendations align closely with the present framework, which achieves coordination through distributed decision-making supported by shared real-time information.

Substantial research has also addressed low-level challenges of physical robot movement and locomotion in real-world environments [22,23]. These mobility issues lie outside the scope of the current study, which focuses primarily on high-level autonomy and multi-agent coordination.

3. RoboCup Rescue Simulation Environment

The Great Hanshin Earthquake struck Kobe, Japan, on 17 January 1995, claiming over 6000 lives and causing more than USD 100 billion in damage [24,25]. Its severity, exacerbated by high urban population density, exposed critical deficiencies in large-scale disaster response. These lessons directly motivated the creation in 2001 of the RoboCup Rescue Agent Simulation (RCRS) as part of the broader RoboCup initiative [26].

RCRS provides a 2D, partially observable, dynamic, discrete-time, and stochastic multi-agent environment designed for Urban Search and Rescue (USAR) research [27]. The simulation replicates a post-earthquake urban crisis through the following core components:

Kernel operates at the system's core and manages the fundamental rules and mechanics of the simulation. It mediates between all components to ensure seamless execution of the overall process flow.

Blockades arise from collapsed buildings at simulation start and appear as black polygons on roads. They are not generated dynamically afterward. Relevant properties used in this study include the following:

- *Coordinates*: X and Y coordinates of the blockade centroid.
- *Position*: ID of the road entity containing the blockade.
- *RepairCost*: Total effort required to remove the blockade completely.

Agents are autonomous programmable entities that perform rescue operations. Three specialized types exist, displayed as colored circles on the map:

- Police Force agents (blue): Clear road blockades; clearing capacity per cycle is governed by the *RepairRate* property.
- Ambulance Team agents (white): Load injured civilians and transport them to a refuge center.
- Fire Brigade agents (red): Extinguish fires and rescue buried civilians.

Civilians appear as green circles. In addition to position-related properties, they possess *Buriedness* (depth under rubble) and *Damage* (health loss per cycle).

Buildings constitute the primary sites of civilian entrapment. Collapse severity and trapped occupants are fixed at simulation start. PF agents must clear entrance debris before other agents can reach trapped individuals.

Agent Perception is provided by a line-of-sight (LOS) mechanism with a default range of 100 m (reduced by obstructions). All perceived changes are delivered via a **ChangeSet** structure that updates automatically whenever new entities or modifications enter the visibility range. This study extensively uses ChangeSet for low-level decision-making by PF agents (detailed in Section 5).

Communication in disaster environments typically suffers from severe infrastructure degradation [26]. RCRS therefore supports configurable communication models. All experiments in this study adopt the medium setting: three radio channels, each with a 1024 Hz bandwidth, and one voice channel offering a 30 m range and a 256-byte maximum message size. Voice messages are sent using `speak` commands; radio messages use `tell` commands.

These components collectively establish RCRS as a realistic and widely accepted benchmark for research on high-level multi-agent coordination under severe resource and communication constraints.

4. TAEMS Modeling Language

The Task Analysis, Environment Modeling, and Simulation framework provides a quantitative foundation for analyzing, explaining, and predicting performance in multi-agent systems operating under uncertainty and incomplete information [28]. TAEMS enables explicit representation of all feasible plans an agent can pursue, together with probabilistic outcomes of each plan [29].

Tasks constitute the fundamental units of activity and are described hierarchically with ***subtasks***, ultimately decomposing into executable ***methods***. Each method produces outcomes characterized by probabilistic distributions across three dimensions: Quality (Q), Cost (C), and Duration (D).

Quality Accumulation Functions (QAFs) specify how quality aggregates from methods or subtasks to parent tasks. The QAFs employed in this study are as follows:

- **q_seq_sum()**: quality of the parent task equals the sum of qualities from subtasks executed sequentially.
- **q_exactly_one()**: quality of the parent task equals the quality of exactly one successfully completed subtask.

Relationships between tasks and methods capture dependencies. The most relevant relationship for rescue scenarios is the *enables* non-local effect, in which the completion of one task (e.g., Remove Blockade) is required before another task (e.g., Dig Civilians) can begin [30,31].

Figure 1 illustrates a typical enables chain: an Ambulance Team agent can only rescue civilians in Building 1 after a Police Force agent clears the road blockade and, if the civilians are buried, a Fire Brigade agent digs out the civilians. Methods may exhibit outcome uncertainty; for example, the method *Remove Blockade on Road A* yields 10 quality units with 80% probability or 30 quality units with 20% probability.

This research focuses on high-level reasoning for PF agents and applies TAEMS modeling to enable task planning and scheduling within the testbed. As demonstrated in prior work [32,33], hierarchical task decomposition with interdependent relationships [31], combined with quantitative modeling of outcome uncertainties in Quality, Cost, and Duration, supports coordinated collaboration while integrating seamlessly with each agent's local planning and scheduling processes. Through this modeling and reasoning framework, PF agents in the RCRS testbed can effectively prioritize tasks to achieve the optimal combination of outcomes.

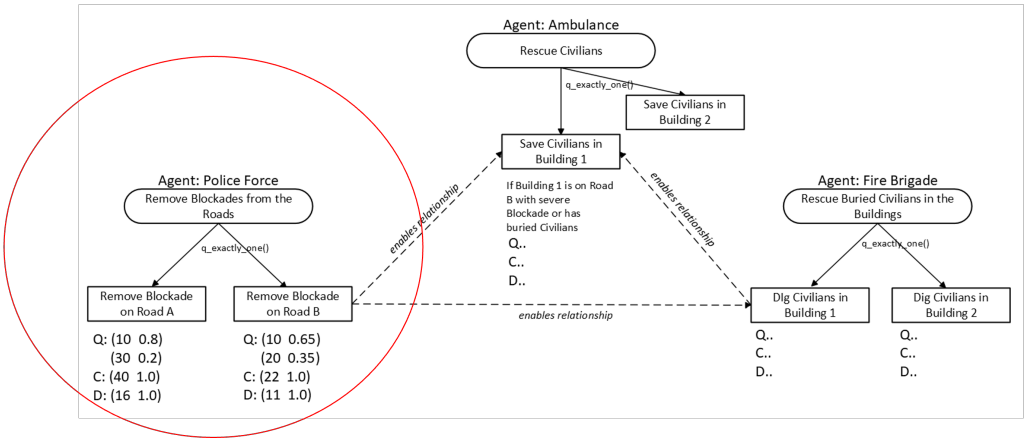


Figure 1. High-level reasoning among different RCRS agents expressed in TAEMS (enables relationships), red circle indicating PF agents with scheduling capability.

5. Methodology

This section describes the integration of TAEMS into the RCRS platform and its use by Police Force agents for real-time high-level task scheduling. Figure 2 depicts the execution cycle of an active PF agent.



Figure 2. Police Force agent's lifecycle.

5.1. Police Force Agent Design

TAEMS is employed exclusively to model high-level blockade-clearing tasks for PF agents. The fixed template shown in Figure 3 serves as the scheduler backbone and is instantiated every cycle with current simulation data.

Each method under the *Path-to-the-Blockade* task node represents one candidate blockade and carries its own Quality, Cost, and Duration (QCD) distributions. The *q_exactly_one()* QAF on this node ensures that exactly one method (i.e., one blockade) is selected as the high-level target for the current cycle.

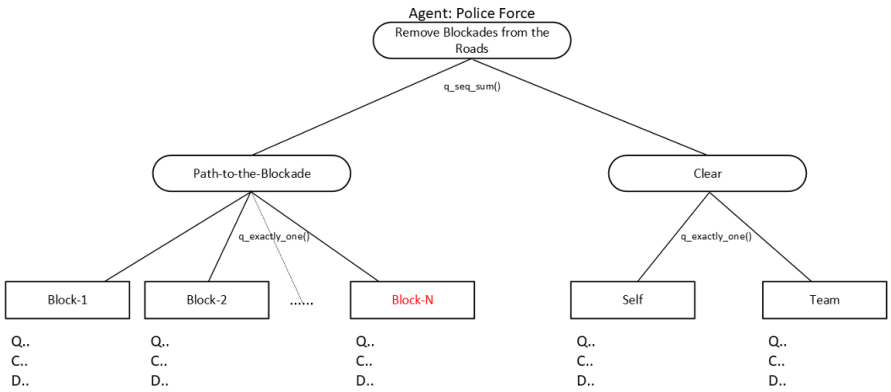


Figure 3. TAEMS tree template for Police Force agents. An arbitrary number N of blocks may be added dynamically.

The template receives blockade information from three sources:

- **Shared Communication Space**: blockades explicitly prioritized by the team (Section 5.3).
- **Heard Voice Messages**: blockade announcements received within the 30 m voice range from other agents.
- **Nearby Blockades**: blockades located on the same road as the PF agent.

Figure 4 illustrates the QCD instantiation process. A baseline quality is assigned to each blockade; this value is multiplied by 3 for blockades near trapped civilians and by 2 for blockades near trapped rescue agents. Duration is computed as follows:

$$\text{Duration} = \frac{\text{RepairCost}}{\text{RepairRate}}$$

where *RepairCost* is the blockade property and *RepairRate* is the PF agent's clearing capacity per cycle.

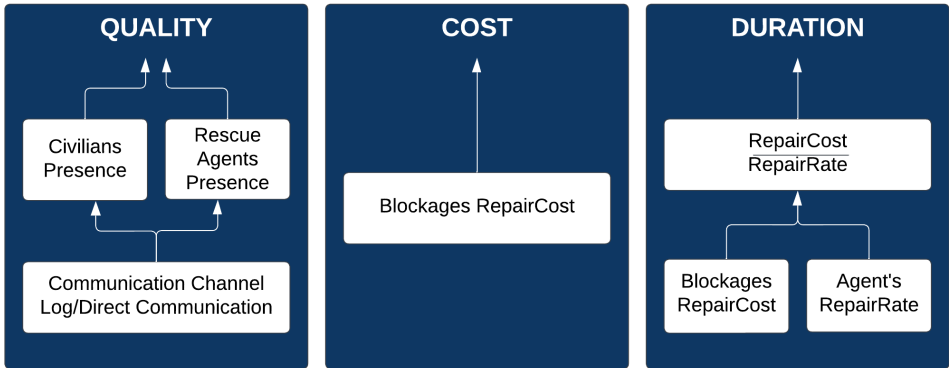


Figure 4. Data feeding from the server to Quality, Cost and Duration.

The selected high-level target may be obstructed by intermediate blockades. These are treated as temporary low-level targets and are identified using the agent's local world model derived from line-of-sight perception (ChangeSet).

5.2. Task Scheduler for PF

Design-to-Criteria (DTC) scheduling [34,35] generates schedules that optimize a weighted combination of objectives. The weights used throughout this study are Quality = 0.6, Duration = 0.3, and Cost = 0.1 [30,33].

At simulation start, each PF agent loads the predefined TAEMS template. Although the template structure is identical for all PF agents, the specific blockades inserted as methods vary according to each agent's current knowledge.

The scheduler proceeds as follows:

1. Extract all candidate methods currently present in the template.
2. Fill in their QCD distributions based on real-time information.
3. Generate a set of alternative schedules S that respect the defined QAFs and intertask relationships (primarily *enables*).
4. Select the schedule $s \in S$ with the highest goodness score according to the DTC criterion Formula (1) .

$$\begin{aligned} goodness(s) = & \frac{Quality(s)}{bestQuality(S)} * QualityWeight \\ & + \frac{maxDuration(S) - Duration(s)}{maxDuration(S)} * DurationWeight \\ & + \frac{maxCost(S) - Cost(s)}{maxCost(S)} * CostWeight \end{aligned} \tag{1}$$

where $bestQuality(S)$, $maxDuration(S)$, and $maxCost(S)$ denote the highest quality, the longest duration, and the highest cost among all schedules in the set of alternative schedules S, respectively. The number of alternative schedules generated at each invocation of the DTC scheduler is configurable. In the experiments reported in Section 6, this value is fixed at 10.

5.3. Control Flows for Agent Action and Interaction

Each agent type follows a dedicated control flow that governs both local actions and interaction with teammates via the shared communication space. All transmitted messages consist of a simulation entity object (civilian or blockade) together with its current position and are sent exclusively through the 30 m voice channel using `speak` commands.

Ambulance Team (AT) and Fire Brigade (FB) agents employ nearly symmetric protocols, acting simultaneously as information producers and consumers (Figure 5):

- Upon encountering an unburied (rescued) civilian, an AT agent transports the civilian directly to the nearest refuge.
- Upon encountering a buried civilian, an AT agent broadcasts the civilian entity via the voice channel.
- Upon receiving a voice message from an FB agent announcing a newly rescued civilian, the AT agent navigates to that location. If the path is clear, the civilian is transported to the refuge; otherwise, the obstructing blockade entity is broadcast.

FB agents operate analogously: they clear rubble from buried civilians, broadcast the civilian entity once rescued, and broadcast any obstructing blockade entities.

Police Force agents function primarily as information consumers (Figure 6). They continuously monitor incoming voice messages and the shared communication space for reported blockade entities, incorporate these entities into their TAEMS schedule template, and proceed to clear the highest-priority ones.

This voice-only, fully decentralized protocol eliminates reliance on radio channels while enabling effective role-specific coordination across heterogeneous teams.

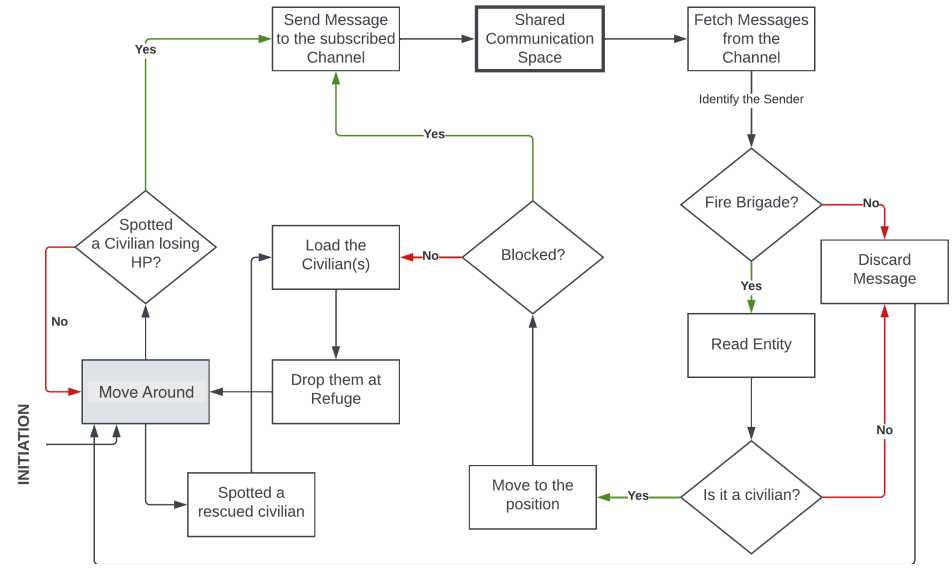


Figure 5. Action and interaction flow for Ambulance Team agents.

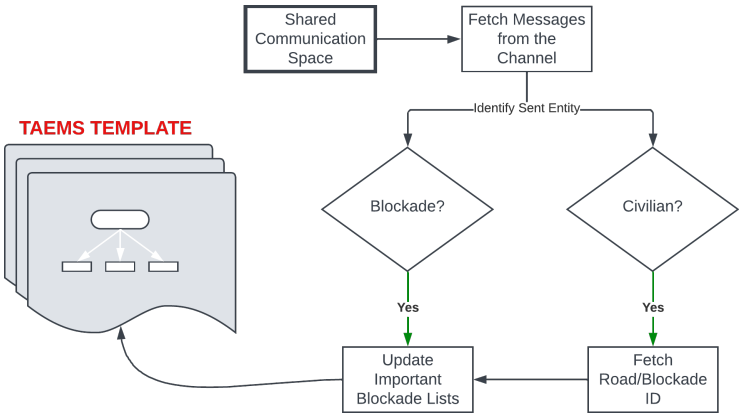


Figure 6. Action and interaction flow for Police Force agents.

6. Experiments and Results

All experiments were conducted on a 2020 MacBook Air (M1 chipset, 8-core CPU, 8-core GPU, 8 GB RAM). Maps and initial configurations were created using the RCRS Map Scenario Editor (V2.0). A single refuge was placed in the bottom-right corner of each map; no other special buildings were used.

6.1. Robustness Experiment

The first experiment evaluates the consistency and reliability of the proposed approach. The scenario consists of one Police Force (PF), one Ambulance Team (AT), and one Fire Brigade (FB) agent together with four civilians (Figure 7). To stress the coordination component, the AT and FB agents were positioned far from the PF agent. The simulation horizon was extended from the default 300 cycles to 500 cycles.

Each configuration (default agents vs. modeled agents) was executed three times on the identical deterministic map. Because the RCRS simulator, map layout, agent positions, and both agent implementations are fully deterministic, differences between runs arise solely from minor operating-system thread-scheduling variations. Results are summarized in Table 1.

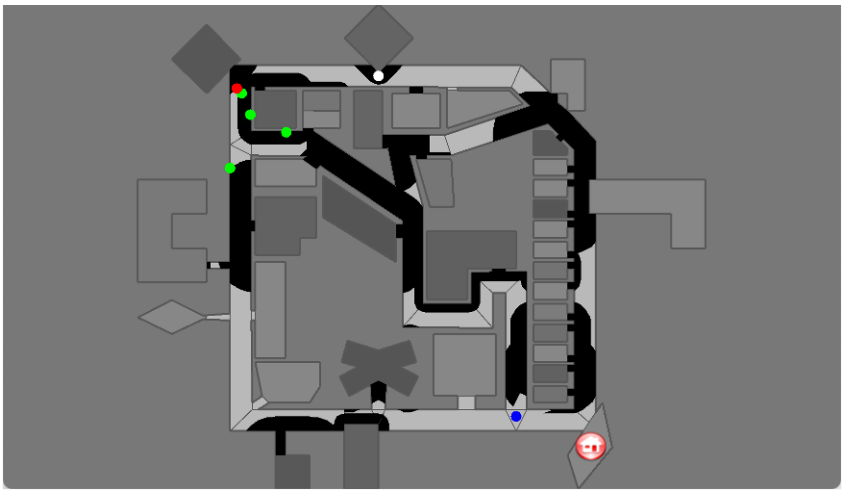


Figure 7. Simulation map used in robustness experiment.

Table 1. Robustness experiment: default vs. modeled agents (AVG: average; CV: coefficient of variation).

Task Completion	Default PF					Modeled PF				
	Run 1	**Run 2**	**Run 3**	**AVG**	**CV**	**Run 1**	**Run 2**	**Run 3**	**AVG**	**CV**
Contact with First Civilian	273	272	273	272.7	0.2%	202	202	203	202.3	0.3%
Contact with Fire Brigade	319	318	319	318.7	0.2%	221	220	221	220.7	0.3%
Contact with Ambulance	340	341	342	341.0	0.3%	259	258	259	258.7	0.2%
Rescued All Civilians	341	341	342	341.3	0.2%	337	337	337	337.0	0%

The coefficient of variation (CV) across the three runs remains below 0.3% for all metrics in both configurations, confirming high reproducibility. The proposed modeled agents consistently achieve earlier first contact with other entities and faster completion of all rescue tasks, demonstrating stable superiority over the default implementation.

In every run, both agent implementations exhibited near-identical behavior across repetitions. The default PF agent (without the proposed task scheduler or communication protocol) established first contact with another human entity at 272–273 timesteps. The proposed PF agent achieved the same milestone at 202–203 timesteps, corresponding to a reduction of approximately 25%.

Comparison of key milestone progression across the six runs reveals that the modeled PF agent consistently achieves earlier contacts (with the first civilian, Fire Brigade, and Ambulance Team), and delivers faster assistance to rescue all civilians. The final time required to deliver all civilians to the refuge shows smaller improvement, primarily because rescued civilians, once unloaded by an AT agent, occasionally fail to autonomously navigate remaining open paths to the refuge—a known limitation of the default civilian movement model in RCRS rather than of the proposed coordination framework.

6.2. Score Comparison Experiment

The second experiment evaluates overall team performance through the final simulation score obtained with default agents versus the proposed agents. Fire and ignition simulators were disabled, so the standard RCRS scoring function simplifies to the following:

$$\text{Score} = (\text{Number of civilians alive}) + (\text{Average health proportion of all civilians}).$$

The health proportion component has a maximum value of 1, yielding a theoretical maximum score of 10 when nine civilians are present.

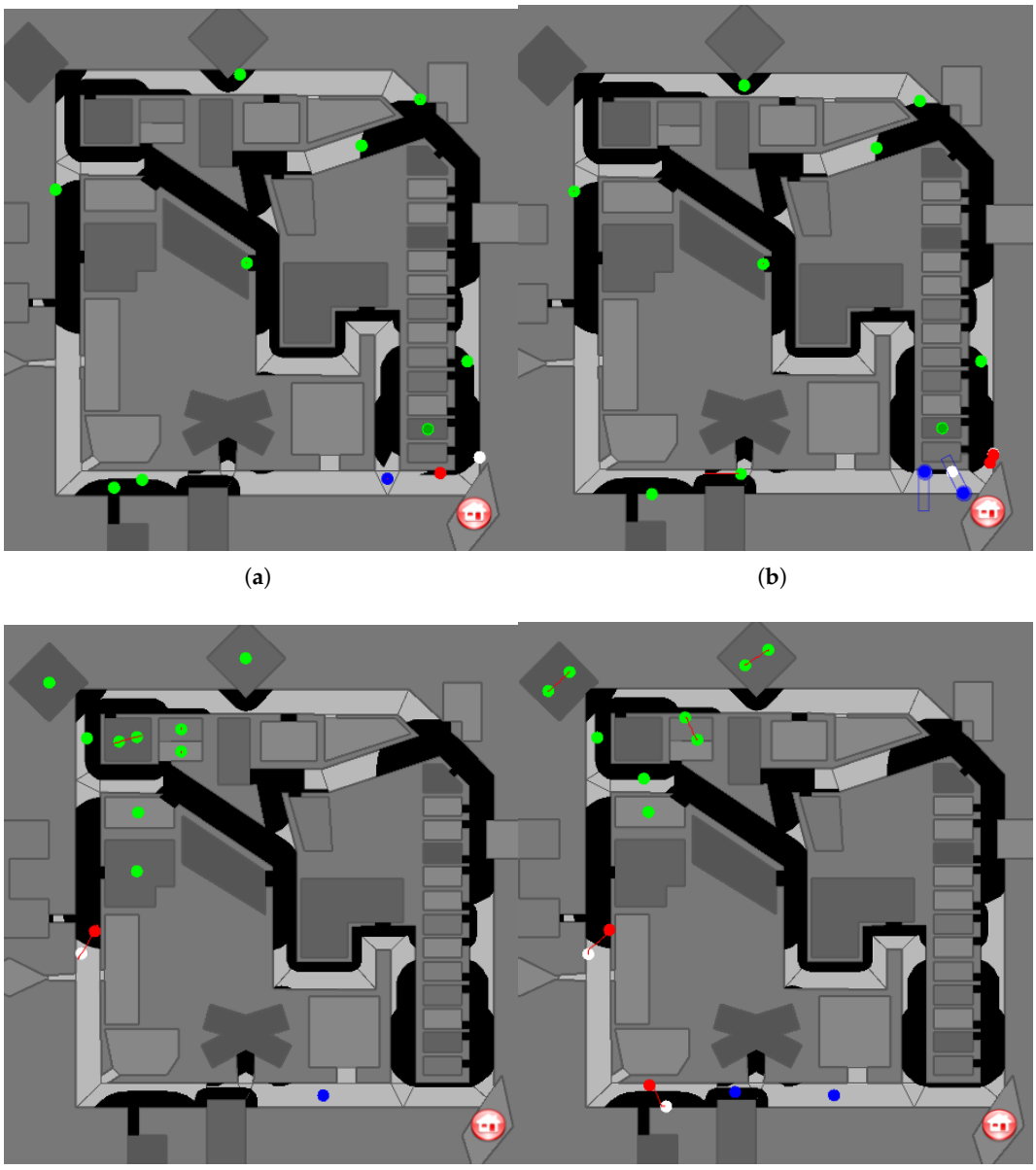
Four configurations were designed to vary the spatial distribution of agents and civilians, thereby evaluating coordination effectiveness under both homogeneous and heterogeneous teams. The scenarios are as follows:

- **Configuration 1 (Figure 8a)**: Nine civilians scattered across the map, with one agent of each type (PF, AT, FB) placed nearby. This layout is relatively straightforward even for default agents.
- **Configuration 2 (Figure 8b)**: Same civilian distribution as Configuration 1, but with two agents of each type. This setting tests coordination among multiple homogeneous agents of the same role.
- **Configuration 3 (Figure 8c)**: Nine civilians clustered together, with one agent of each type positioned far apart. This configuration emphasizes long-range communication-driven coordination, particularly the ability of the distant PF agent to prioritize relevant blockades early.
- **Configuration 4 (Figure 8d)**: Same civilian clustering as Configuration 3, but with two agents of each type. The increased travel distances cause civilians to accumulate more damage before rescue, raising overall difficulty.

A performance comparison between default and proposed agents across the four configurations is presented in Table 2 and Figure 9. Each reported score corresponds to a single deterministic run; owing to the fully deterministic nature of the RCRS simulator, map, initial positions, and agent implementations, identical configurations always yield identical outcomes.

Table 2. Simulation score results for different configurations.

No.	Agent Distribution	Civilian Distribution	Humans	Scores (Out of 10)		Improvement
				Default Agents	Modeled Agents	
1	Nearby	Scattered	1PF/1AT/1FB/9civilians	8.889	9.960	12.05%
2	Nearby	Scattered	2PF/2AT/2FB/9civilians	8.889	9.995	12.44%
3	Scattered	Nearby	1PF/1AT/1FB/9civilians	5.556	7.644	37.58%
4	Scattered	Nearby	2PF/2AT/2FB/9civilians	4.356	6.455	48.19%



(**a**) (**b**)

(**c**) (**d**)

Figure 8. Four simulation maps: (**a**) Config1: scattered civilians with nearby rescue agents (1 PF, 1 AT, 1 FB) (**b**) Config 2: scattered civilians with nearby rescue agents (2 PF, 2 AT, 2 FB). (**c**) Config 3: nearby civilians with scattered rescue agents (1 PF, 1 AT, 1 FB). (**d**) Config 4: nearby civilians with scattered rescue agents (2 PF, 2 AT, 2 FB).

The proposed agents consistently outperform the default agents in all configurations, with score improvements ranging from 12% to 48%. The largest gains appear in Configurations 3 and 4, where civilians are tightly clustered while rescue agents start widely dispersed. In these scenarios, default agents lack mechanisms to identify and prioritize globally critical blockades, resulting in prolonged civilian damage accumulation.

These results demonstrate the effectiveness of the proposed coordination framework: even without prior knowledge of teammate or civilian locations, PF agents—through TAEMS-based scheduling and real-time voice-channel information sharing—systematically select blockades that maximize overall team rescue outcome.

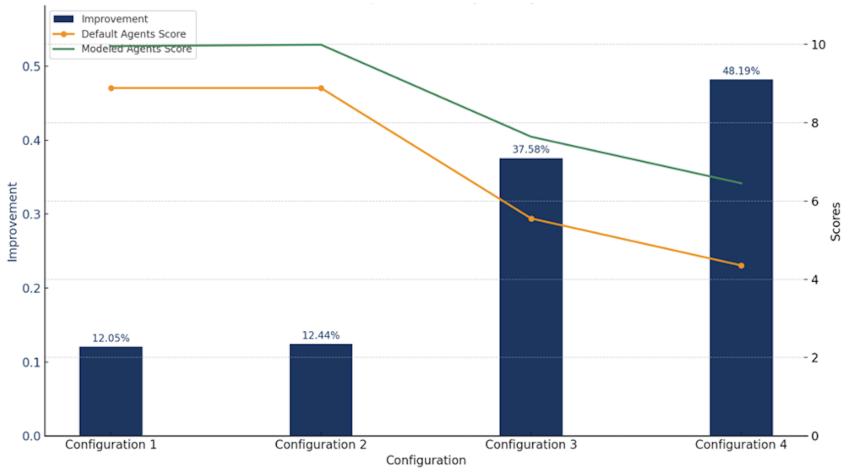


Figure 9. Comparison of all 4 configurations (orange line: default agent scores; green line: modeled agent scores).

7. Conclusions and Future Work

This study addressed coordinated task execution in the RCRS environment by integrating the TAEMS framework with a lightweight voice-only communication protocol. The resulting coordination layer was incorporated into the standard RCRS agent architecture without modifying the underlying simulator.

Extensive evaluation across diverse scenarios demonstrated that the proposed approach consistently improves team performance by 12% to 48% over default agents. The largest gains occur in configurations where civilians are clustered and rescue agents start widely separated—situations in which default agents cannot effectively identify globally critical blockades. These results confirm that structured task modeling combined with real-time information sharing enables robust decentralized coordination among heterogeneous agents under severe communication constraints.

Several directions for future work remain. Currently, full TAEMS modeling and DTC scheduling are applied only to Police Force agents. Extending these capabilities to Ambulance Team and Fire Brigade agents would provide uniform decision-making sophistication across the team. Allowing agents to selectively broadcast commitments and introducing priority-aware message selection would further reduce channel congestion and enhance allocation efficiency.

Transitioning to physical multi-robot systems requires addressing additional challenges. Accurate local maps cannot be assumed; agents must perform simultaneous exploration, collaborative mapping, and incremental model refinement. Initial travel-time estimates can use conservative geometric heuristics, progressively replaced by empirically updated values from shared perception. The probabilistic QCD distributions native to TAEMS are currently underutilized and should be leveraged to handle real-world outcome variability.

Physical platforms impose further constraints: tight integration of sensing and locomotion for coverage guarantees, robust communication with acknowledgments and retransmissions, energy-aware scheduling, and adherence to hard real-time deadlines. Fault-tolerance mechanisms—dynamic role reassignment and load balancing in response to sensor failure, actuator damage, or battery depletion—are also essential to maintain team performance when individual robots degrade.

Addressing these extensions forms the primary roadmap for evolving the present simulation-validated framework toward reliable deployment in real-world disaster response operations.

Author Contributions: Conceptualization, S.Z. and E.C.; methodology, S.Z., A.B. and E.C.; software, A.B.; validation, A.B. and E.C.; writing—original draft preparation, A.B.; writing—review and editing, S.Z.; visualization, A.B.; supervision, S.Z.; project administration, S.Z.; funding acquisition, S.Z. and E.C. All authors have read and agreed to the published version of the manuscript.

Funding: This material is based upon work supported by the Office of Naval Research (ONR) through the contract N00014-21-1-2236 with UMass Dartmouth. Any opinions, findings, conclusions, or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of ONR or the U.S. Government.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

Acknowledgments: The authors express gratitude to David K. Degbor for developing the communication methods among agents, which were adapted and refined for use in this study. The authors also acknowledge Paul Naylor for his valuable support with computational and laboratory resources.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Saur, D.; Geihs, K. IMPERA: Integrated Mission Planning for Multi-Robot Systems. *Robotics* **2015**, *4*, 435–463. [CrossRef]
2. Manjanna, S.; Dudek, G.; Hansen, J.; Pizarro, O.; Zykov, V. Informed Scientific Sampling in Large-scale Outdoor Environments. In Proceedings of the International Conference on Intelligent Robots and Systems (IROS), Macau, China, 4–8 November 2019.
3. Murphy, R.R. Trial by fire [rescue robots]. *IEEE Robot. Autom. Mag.* **2004**, *11*, 50–61. [CrossRef]
4. Akin, H.L.; Ito, N.; Jacoff, A.; Kleiner, A.; Pellenz, J.; Visser, A. Robocup rescue robot and simulation leagues. *AI Mag.* **2012**, *34*, 78. [CrossRef]
5. Verma, J.K.; Ranga, V. Multi-robot coordination analysis, taxonomy, challenges and future scope. *J. Intell. Robot. Syst.* **2021**, *102*, 10. [CrossRef] [PubMed]
6. Almadhoun, R.; Taha, T.; Seneviratne, L.; Zweiri, Y. A survey on multi-robot coverage path planning for model reconstruction and mapping. *SN Appl. Sci.* **2019**, *1*, 847. [CrossRef]
7. Yan, Z.; Jouandeau, N.; Cherif, A.A. A survey and analysis of multi-robot coordination. *Int. J. Adv. Robot. Syst.* **2013**, *10*, 399. [CrossRef]
8. Su, X.; Zhang, M.; Bai, Q. Dynamic task allocation for heterogeneous agents in disaster environments under time, space and communication constraints. *Comput. J.* **2015**, *58*, 1776–1791. [CrossRef]
9. RoboCup Rescue Simulation. Available online: https://rescuesim.robocup.org (accessed on 5 October 2025).
10. Horling, B.; Lesser, V.; Vincent, R.; Wagner, T.; Raja, A.; Zhang, S.; Decker, K.; Garvey, A. The TAEMS White Paper. 1999. Available online: http://mas.cs.umass.edu/paper/182 (accessed on 2 December 2023).
11. Degbor, D.; Bedi, A.; Zhang, S.; Chabot, E. Communication-Facilitated Coordination in Agent Team Rescue Mission. In Proceedings of the 2023 Congress in Computer Science, Computer Engineering & Applied Computing (CSCE), Las Vegas, NV, USA, 24–27 July 2023; pp. 19–22. [CrossRef]
12. Lu, J.; Liang, Z.; Li, X.; Zhu, Z.; Wei, B.; Fang, F. The Application of Adaptive Ant-colony A Hybrid Algorithm Based on Objective Evaluation Factor in RoboCup Rescue Simulation Dynamic Path Planning. In *IOP Conference Series: Materials Science and Engineering*; IOP Publishing: Bristol, UK, 2019; Volume 631, p. 052028. [CrossRef]
13. Lyu, Q.; Cao, X.; Yu, Y.; Xue, S.; Yang, J.; Wu, J.; Beheshti, A. A Multiagent-based Framework on RoboCup Simulation System for Enhancing Rescue Operation during Dynamic Disaster Environments. In *Companion Proceedings of the ACM on Web Conference 2025*; Association for Computing Machinery: New York, NY, USA, 2025; WWW'25; pp. 1180–1183. [CrossRef]
14. Ando, K.; Maeda, R.; Uehara, H.; Fujisawa, J.; Matsunaga, I.; Suzuki, R.; Kato, K.; Shimada, Y.; Fujii, S.; Uchitane, T.; et al. Designing a Rescue Strategy Emphasizing Distributed Control in RoboCupRescue Simulation: AIT-Rescue: RoboCup 2024 Rescue Simulation League Champion. In *Robot World Cup*; Springer: Berlin/Heidelberg, Germany, 2024; pp. 436–447.
15. Sedaghat, M.N.; Nejad, L.P.; Iravanian, S.; Rafiee, E. Task allocation for the police force agents in RoboCup Rescue simulation. In *Robot Soccer World Cup*; Springer: Berlin/Heidelberg, Germany, 2006; Volume 4020, pp. 656–664.

16. Pujol-Gonzalez, M.; Cerquides, J.; Farinelli, A.; Meseguer, P.; Rodríguez-Aguilar, J.A. Binary Max-Sum for Multi-Team Task Allocation in RoboCup Rescue. In Proceedings of the International Conference on Autonomous Agents and Multi-Agent Systems, Paris, France, 5–9 May 2014.
17. Ferreira, P.R., Jr.; dos Santos, F.; Bazzan, A.L.C.; Epstein, D.; Waskow, S.J. RoboCup Rescue as multiagent task allocation among teams: Experiments with task interdependencies. *J. Auton. Agents-Multi-Agent Syst.* **2010**, *20*, 421–443. [CrossRef]
18. Ramchurn, S.; Farinelli, A.; Macarthur, K.; Jennings, N. Decentralized Coordination in RoboCup Rescue. *Comput. J.* **2010**, *53*, 1447–1461. [CrossRef]
19. Drew, D.S. Multi-Agent Systems for Search and Rescue Applications. *Curr. Robot. Rep.* **2021**, *2*, 189–200. [CrossRef]
20. Xueke, Y.; Yu, Z.; Junren, L.; Kaiqiang, W. Multi-agent Task Coordination Method Based on RCRS. In *2021 International Conference on Autonomous Unmanned Systems (ICAUS 2021)*; Wu, M., Niu, Y., Gu, M., Cheng, J., Eds.; Springer: Singapore, 2022; pp. 2582–2593.
21. Sun, L.; Yang, Y.; Duan, Q.; Shi, Y.; Lyu, C.; Chang, Y.C.; Lin, C.T.; Shen, Y. Multi-Agent Coordination across Diverse Applications: A Survey. *arXiv* **2025**, arXiv:2502.14743. [CrossRef]
22. Rybczak, M.; Popowniak, N.; Lazarowska, A. A Survey of Machine Learning Approaches for Mobile Robot Control. *Robotics* **2024**, *13*, 12. [CrossRef]
23. Chen, L.; Ye, S.; Sun, C.; Zhang, A.; Deng, G.; Liao, T. Optimized Foothold Planning and Posture Searching for Energy-Efficient Quadruped Locomotion over Challenging Terrains. In Proceedings of the 2020 IEEE International Conference on Robotics and Automation (ICRA), Paris, France, 31 May–31 August 2020; pp. 399–405. [CrossRef]
24. CE: Kobe Earthquake. 2023. Available online: https://www.nationalgeographic.org/thisday/jan17/kobe-earthquake/ (accessed on 2 December 2023).
25. Encyclopaedia Britannica. Kobe Earthquake of 1995. 2022. Available online: https://www.britannica.com/event/Kobe-earthquake-of-1995 (accessed on 2 December 2023).
26. Kitano, H.; Tadokoro, S. RoboCup rescue: A grand challenge for multiagent and intelligent systems. *AI Mag.* **2001**, *22*, 39.
27. RoboCup Rescue Simulation Server Team. RoboCup Rescue Simulation Server Manual. 2023. Available online: https://roborescue.github.io/rcrs-server/rcrs-server/rcrs-server-manual.pdf (accessed on 2 December 2023).
28. Decker, K.S.; Lesser, V.R. Quantitative modeling of complex environments. *Int. J. Intell. Syst. Account. Financ. Managt.* **1993**, *2*, 215–234. [CrossRef]
29. So, Y.; Durfee, E.H. Designing tree-structured organizations for computational agents. *Comput. Math. Organ. Theory* **1998**, *4*, 295–315. [CrossRef]
30. Horling, B.; Lesser, V. A survey of multi-agent organizational paradigms. *Knowl. Eng. Rev.* **2005**, *19*, 281–316. [CrossRef]
31. Becker, M.; Lesser, V.; Zilberstein, S. Analyzing Myopic Approaches for Multi-Agent Communication. *Comput. Intell.* **2003**, *19*, 179–202.
32. Lesser, V. Reflections on the Nature of Multi-Agent Coordination and Its Implications for an Agent Architecture. *Auton. Agents-Multi-Agent Syst.* **2012**, *26*, 35–57. [CrossRef]
33. Lesser, V.; Decker, K.; Wagner, T.; Carver, N.; Garvey, A.; Horling, B.; Neiman, D.; Podorozhny, R.; Prasad, M.N.; Raja, A.; et al. Evolution of the GPGP/TAEMS Domain-Independent Coordination Framework. *Auton. Agents-Multi-Agent Syst.* **2004**, *9*, 87–143. [CrossRef]
34. Decker, K.; Lesser, V. Designing a family of coordination algorithms. In Proceedings of the First International Conference on Multi-Agent Systems (ICMAS-95), San Francisco, CA, USA, 12–14 June 1995; pp. 73–80.
35. Wagner, T.; Garvey, A.; Lesser, V. Criteria-directed task scheduling. *Int. J. Approx. Reason.* **1998**, *19*, 91–118. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.